

PLATFORM AS A SERVICE

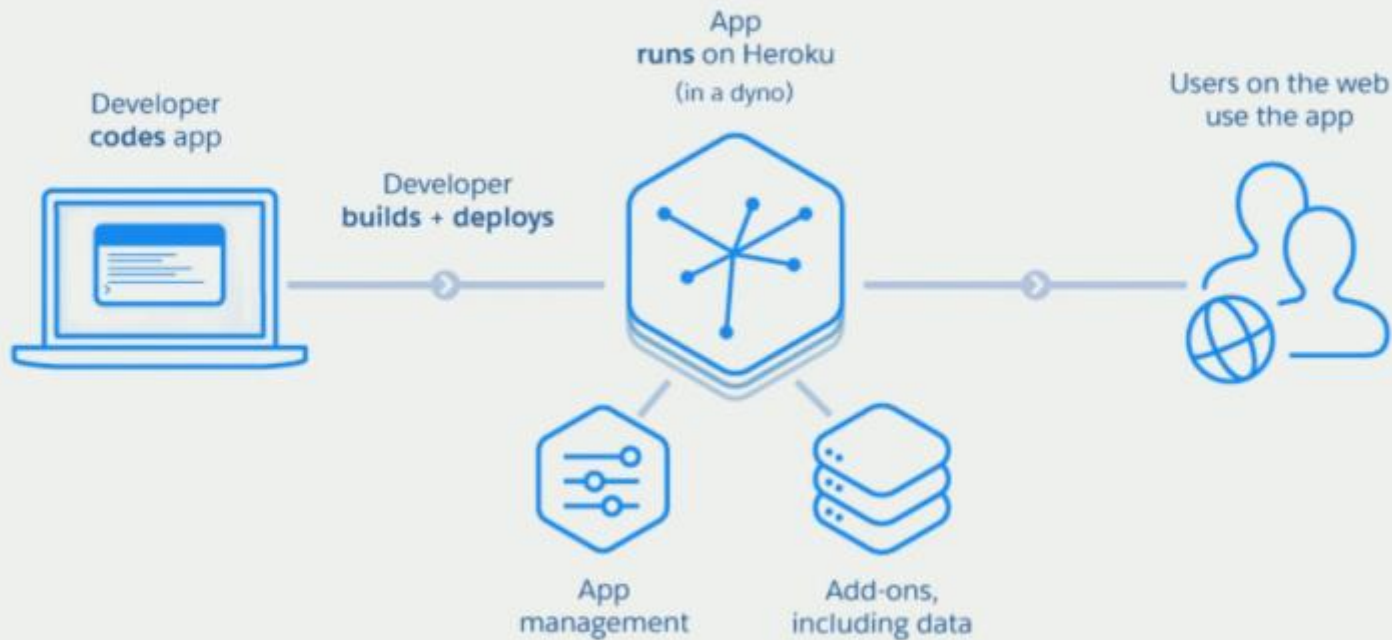
Mr. Pathan Bilalkhan R.

Assistant Professor

Computer Science & Engineering Department

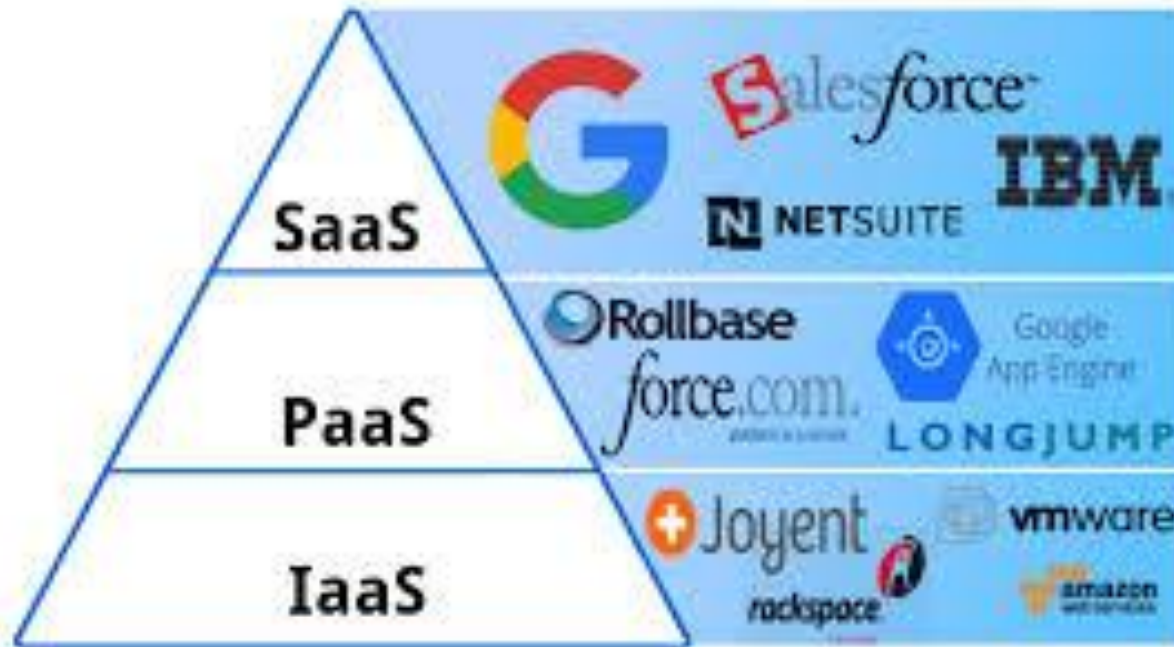
Platform as a Service (PaaS)

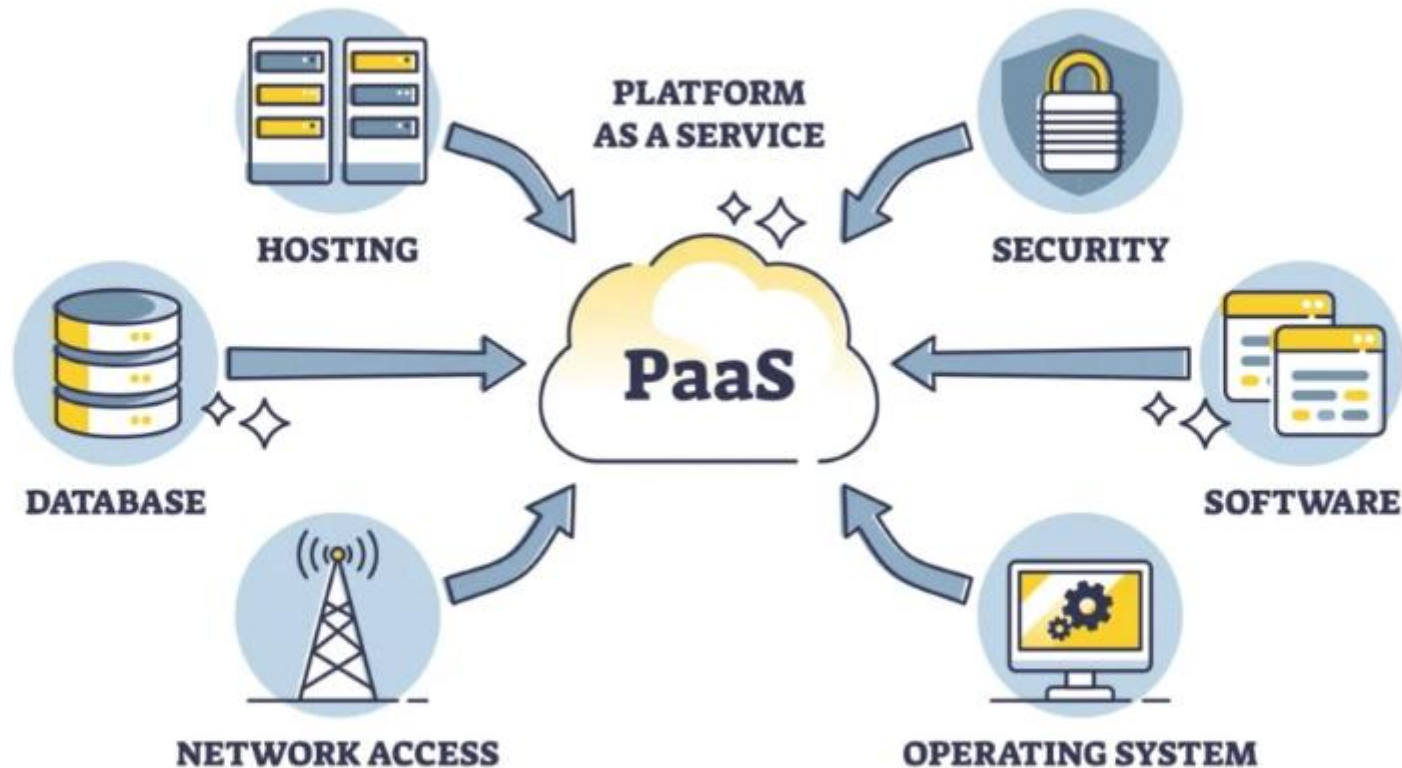
What is PaaS?



What is PaaS?

- Platform to build, deploy, and manage applications
- No need to manage physical servers or OS
- Focus on code and application logic





PaaS Architecture Layers



Software as a Service | SaaS

SaaS is also known as "**On-Demand Software**". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.

SaaS Services



Popular SaaS Providers



Real-World Use Cases

- Online education platforms
- Banking & financial dashboards
- Social networking apps
- Healthcare applications
- IoT backends

Application Lifecycle in PaaS

Plan

Code

Build

Test

Deploy

Monitor

Advantages of PaaS



Reduces the coding time

Development capabilities

Available at Multiple platforms

Tools at economical costs

Regular management of appliance lifecycle

Disadvantages of PaaS

While PaaS provides convenience, scalability, and faster development, it also comes with few drawbacks that every business and developer should consider.

1. Limited Control Over Infrastructure
2. Vendor Lock-In
3. Security and Compliance Risks
4. Performance Issues and Downtime

CLOUD PLATFORM & MANAGEMENT

Roles in PaaS Ecosystem

Role

Cloud Provider

Developer

Administrator

End Users

Responsibility

Offer platform & services

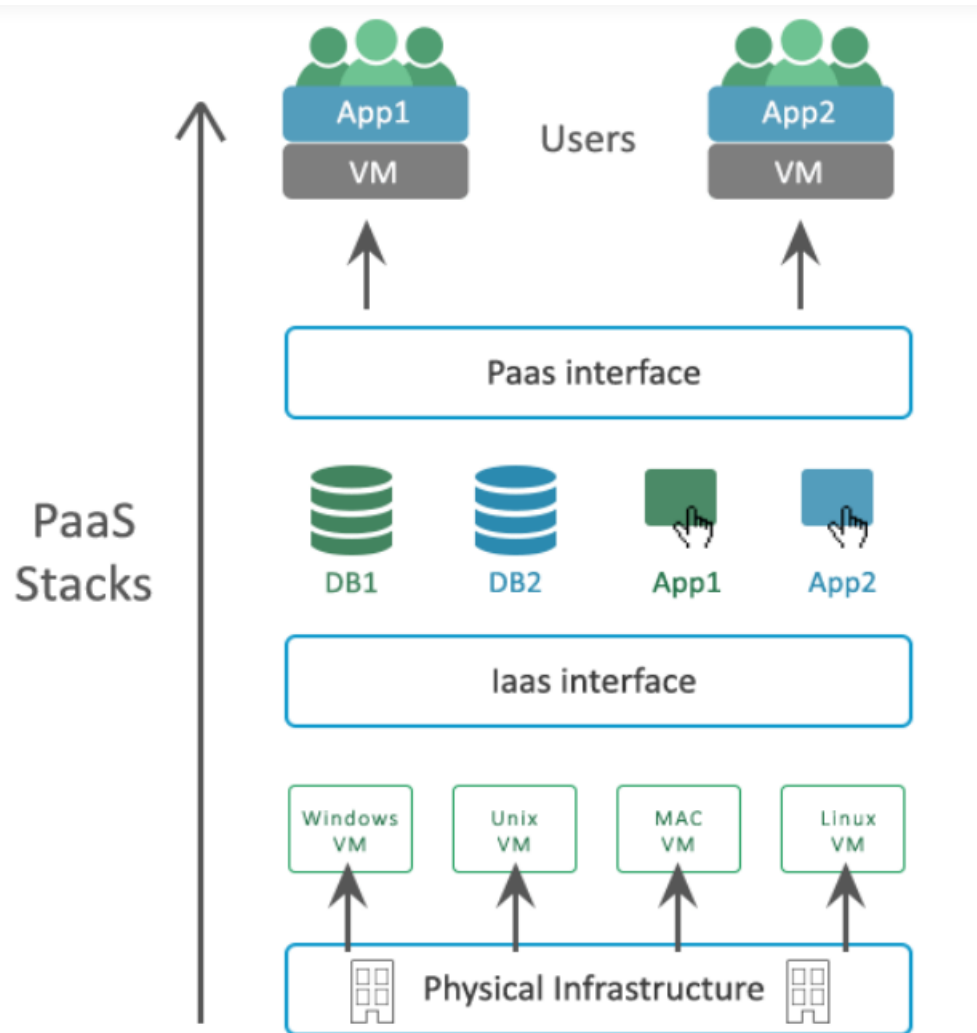
Build & deploy apps

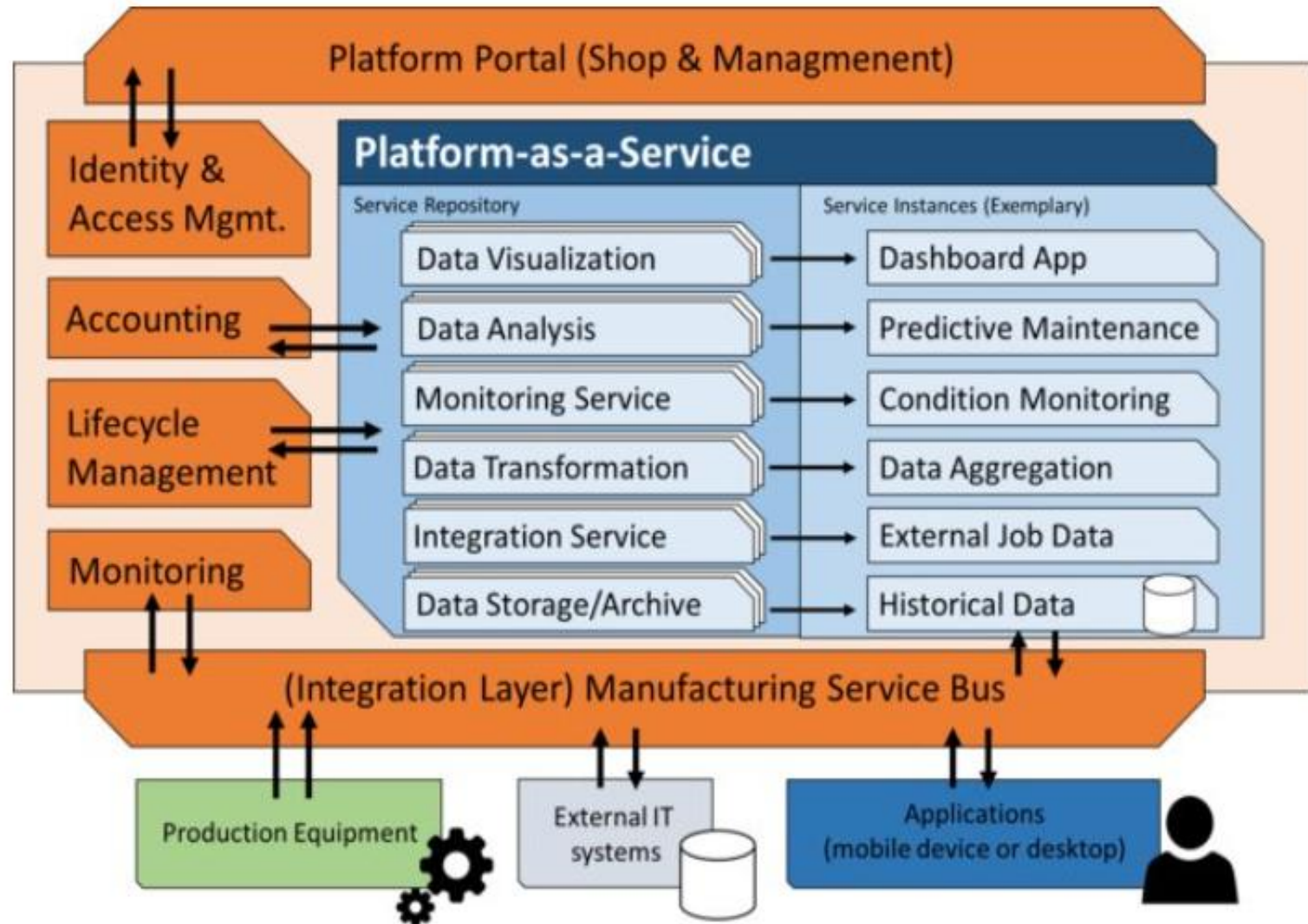
Manage users & resources

Use final application

PaaS ARCHITECTURE & COMPONENTS

- PaaS is not an alternative for the existing IT infrastructure of a business. Instead, a business utilizes the PaaS platform for working with important services like Java development or application hosting.
- A PaaS provider ensures that a resilient platform is provided to the users that can be used for installing applications and data sets. With the help of the PaaS platform, users can completely concentrate on creating and running the applications instead of building the underlying infrastructure and maintaining it.





PaaS and DevOps

PaaS supports:

- ✓ Continuous Integration
- ✓ Continuous Deployment
- ✓ Automation
- ✓ Git-based workflows

EXAMPLES & CASE STUDIES

Platform as a Service (PaaS) provides developers with a cloud-based environment to build, host, and manage applications without the complexity of maintaining underlying infrastructure.

Popular PaaS Examples (2026)

- **AWS Elastic Beanstalk:** A service where developers upload code and the platform automatically handles deployment, capacity provisioning, and load balancing.
- **Google App Engine:** A highly scalable environment for developing and hosting web applications using languages like Java or Python.
- **Heroku:** A developer-centric platform that simplifies the entire application lifecycle, from deployment to scaling.
- **Microsoft Azure App Service:** Provides tools to create, host, and manage RESTful APIs and mobile app backends.
- **Adobe Commerce (formerly Magento):** A specialized PaaS for building and managing e-commerce experiences.
- **Red Hat OpenShift:** A Kubernetes-powered platform designed for managing containers and automating CI/CD workflows.

Key Use Cases

- **API Development:** Platforms like **IBM Cloud Foundry** provide dedicated tools for hosting and managing secure APIs for mobile and web apps.
- **Microservices Architectures:** PaaS supports the deployment of modular, event-driven architectures that can scale independently.
- **DevOps & CI/CD Automation:** Integrates with tools like **GitLab** to automate testing and code deployment to staging environments.
- **Legacy Application Migration:** Organizations use PaaS to modernize expensive legacy applications, potentially reducing operational costs by 50-90%.

Case Studies

- **Product Hunt (via Heroku):** Used the Heroku platform to scale rapidly from a startup to a major platform by focusing on product development rather than server maintenance.
- **Supplier Negotiation (Generic Enterprise):** A company used PaaS to streamline supplier negotiations by building custom tools for mass negotiation line uploads and real-time attribute editing without buying in-house hardware.
- **Medical Device & E-Bike Manufacturers (via Product-as-a-Service):** Several manufacturers used PaaS infrastructure to shift from traditional sales to subscription-based models, speeding up ROI through standardized cloud environments.

Real-World Use Cases

- ✓ Online learning portals
- ✓ Ticket booking systems
- ✓ Banking apps
- ✓ IoT backends
- ✓ Social media apps
- ✓ E-commerce backends

When NOT to use PaaS

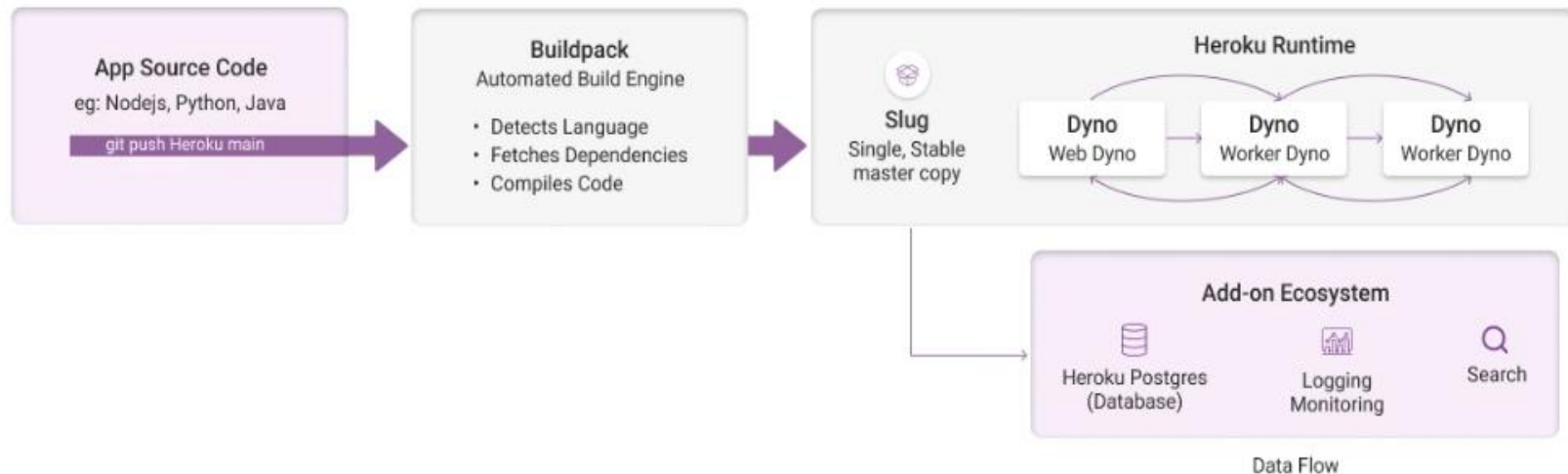
- ✗ When full control over OS needed
- ✗ When using legacy systems
- ✗ When compliance requires on-prem infrastructure

HOW TO IMPLEMENT PaaS

Steps to implement PaaS in organization:

- 1** Choose infrastructure (cloud or on-prem)
- 2** Install virtualization layer
- 3** Install platform layer
- 4** Configure runtime environments
- 5** Create developer environments
- 6** Set up deployment pipelines
- 7** Monitor & secure platform

The Heroku Application Lifecycle



Feature	IaaS	PaaS	SaaS
Definition	Provides virtualized computing resources like servers, storage, and networking.	Offers a platform with tools and environments for application development.	Delivers ready-to-use software applications over the internet.
Control Level	High: Users manage OS, middleware, apps, and data.	Medium: Users control apps and data; the provider manages infrastructure.	Low: Users only manage software configuration and usage.
Examples	AWS EC2 , Microsoft Azure VM , Google Compute Engine .	AWS Elastic Beanstalk , Google App Engine, Heroku.	Google Workspace, Salesforce, Dropbox.
Target Users	IT administrators, developers requiring full control of infrastructure.	Developers looking for a managed platform to build and deploy applications.	End-users needing ready-to-use applications without technical expertise.
Use Cases	Hosting websites, storage, disaster recovery, virtual machines.	Software development, app testing, and deployment.	Email, CRM , file sharing, and collaboration tools.
Infrastructure Access	Provides direct access to virtualized hardware.	Abstracts the infrastructure, offering tools and frameworks.	No access to underlying infrastructure.

Parul[®]
University

NAAC
GRADE **A++**



<https://paruluniversity.ac.in/>

